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Task 2(HPC)
Level: Third Year
Department: AI (General)

# Cash Time

| Matrix Size (N×N) | Traversal Type | Real Time (real) | User Time (user) | Sys Time (sys) |
|---|---|---|---|---|
| 100 | Row | 0m0.001s | 0m0.001s | 0m0.000s |
| 100 | Col | 0m0.002s | 0m0.000s | 0m0.001s |
| 1000 | Row | 0m0.007s | 0m0.007s | 0m0.000s |
| 1000 | Col | 0m0.007s | 0m0.003s | 0m0.004s |
| 10000 | Row | 0m0.823s | 0m0.526s | 0m0.297s |
| 10000 | Col | 1m1.560s | 1m1.200s | 0m0.360s |

Real Time (real): Total elapsed clock time from start to finish.

User Time (user): CPU time spent executing your program's code.

System Time (sys): CPU time spent in the OS kernel

```
rana@DESKTOP-61PG282:~$ time ./row

real    0m0.001s
user    0m0.001s
sys     0m0.000s
rana@DESKTOP-61PG282:~$ nano col.c
rana@DESKTOP-61PG282:~$ gcc -O0 col.c -o col
rana@DESKTOP-61PG282:~$ time ./col

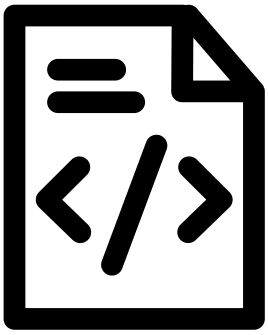
real    0m0.002s
user    0m0.000s
sys     0m0.001s
rana@DESKTOP-61PG282:~$ nano row.c
rana@DESKTOP-61PG282:~$ gcc -O0 row.c -o row
rana@DESKTOP-61PG282:~$ time ./row

real    0m0.007s
user    0m0.007s
sys     0m0.000s
rana@DESKTOP-61PG282:~$ nano col.c
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real    0m0.007s
user    0m0.003s
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rana@DESKTOP-61PG282:~$ nano row.c
rana@DESKTOP-61PG282:~$ gcc -O0 row.c -o row
rana@DESKTOP-61PG282:~$ time ./row

real    0m0.823s
user    0m0.526s
sys     0m0.297s
rana@DESKTOP-61PG282:~$  nano col.c
rana@DESKTOP-61PG282:~$  gcc -O0 col.c -o col
rana@DESKTOP-61PG282:~$  time ./col

real    0m1.560s
user    0m1.200s
sys     0m0.360s
rana@DESKTOP-61PG282:~$
```



## Row-by-Row

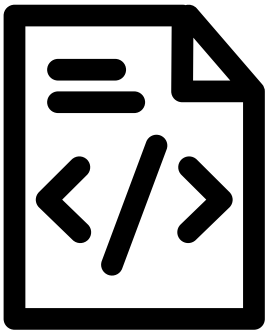
```
#include <stdio.h>
#define N 1000

int main() {
    static int matrix[N][N];
    long long sum = 0;

    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            matrix[i][j] = i + j;
        }
    }

    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            sum += matrix[i][j];
        }
    }

    return 0;
}
```



## Column-by-Column

```
#include <stdio.h>
#define N 1000

int main() {
    static int matrix[N][N];
    long long sum = 0;

    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            matrix[i][j] = i + j;
        }
    }

    for (int j = 0; j < N; j++) {
        for (int i = 0; i < N; i++) {
            sum += matrix[i][j];
        }
    }

    return 0;
}
```